

# MGK-RAG: Multi-Granularity Knowledge Guided Retrieval-Augmented Generation for Radiology Report

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## Abstract

Retrieval-augmented generation (RAG) is an effective approach to enhancing the factual accuracy of radiology reports. However, existing methods primarily model coarse-grained image-report correspondences, ignoring semantic relations among reports that capture hierarchical and fine-grained pathological knowledge. As a result, the learned representations fail to reflect detailed clinical semantics, causing factual inconsistencies in generated reports. Therefore, we propose a multi-granularity knowledge-integrated RAG framework for radiology reports. Specifically, we utilize multi-granularity semantic similarities, derived from the text modality, to adjust the original cross-modal contrastive learning loss. This guides the multimodal retriever to learn a finer-grained clinical semantic alignment. Then, we utilize cross attention to obtain enhanced visual features by integrating the retrieved reports with the original images, thus enhancing the factual accuracy of report generation. The effectiveness of our method was verified on two widely used benchmarks, achieving superior performance in both language generation and key clinical metrics.

## CCS Concepts

- Computing methodologies → Natural language generation;
- Applied computing → Health informatics.

## Keywords

Radiology Report Generation, Large Language Model, Multi-Granularity Knowledge Guided

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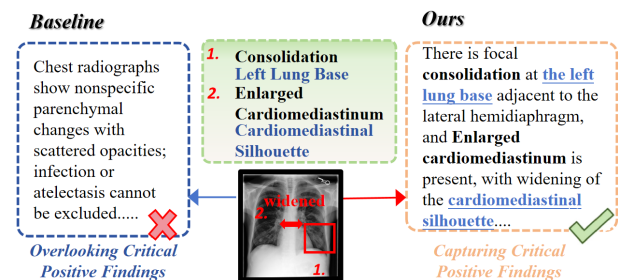


Figure 1: Challenges of existing RAG- and LLM methods: they often produce factually inaccurate or vague outputs.

## 1 Introduction

Automatically generating factually accurate, coherent radiology reports is a core challenge in multimodal medical artificial intelligence, with the potential to reduce radiologist workload and improve interpretation.

Early end-to-end generative models, from CNN-RNN [8] and R2Gen [3] to recent LLM-based architecture [17], have improved report fluency but still produce factually inaccurate reports, which are unacceptable in clinical settings [11, 12]. CXR-RePaiR-Gen [13] first applied RAG to radiology report generation by retrieving similar reports to constrain generation and reduce factual hallucinations. LaB-RAG [14] then introduced a label-boosted retrieval method using predicted pathological labels to guide the process. However, these labels are coarse-grained and cannot effectively distinguish reports that share the same labels but differ in factual details. Subsequently, CBM-RAG [1] introduced a concept bottleneck model to extract finer-grained concepts to guide retrieval, but it neglected the critical, fine-grained entity relationships. Furthermore, while FactMM-RAG [15] leveraged RadGraph to mine factual report pairs for its fact-aware retriever, it does not exploit information at multiple semantic granularities to more precisely guide the alignment process.

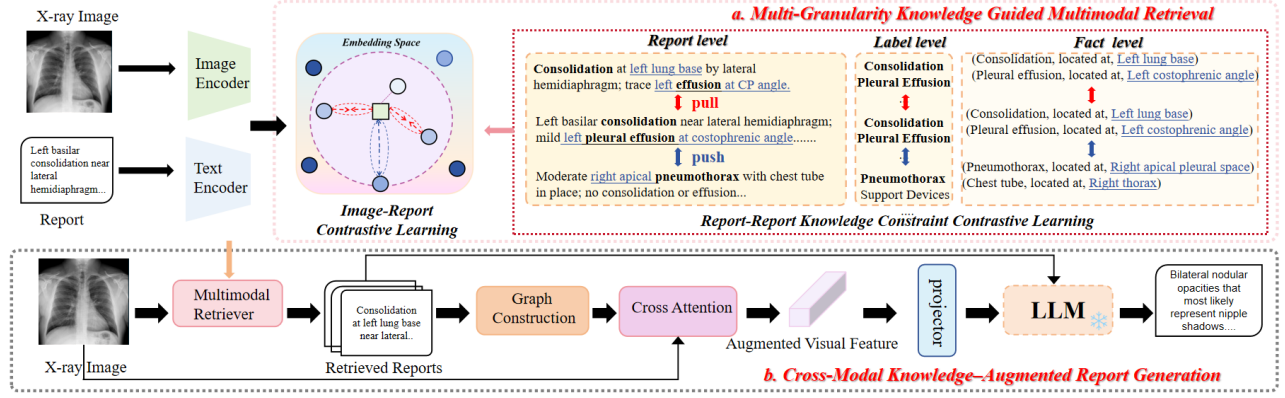


Figure 2: The overall architecture of our proposed method.

Reports from different patients may describe very similar clinical findings, but standard contrastive learning only considers within-pair image-text alignment and ignores this intra-report consistency. We utilize multi-granular intra-report similarities to guide image-text alignment, achieving finer-grained alignment. Fig. 1 illustrates the limitation of alignment lacking intra-modal multi-granularity knowledge constraints. A baseline model without these constraints generates a report that omits key positive findings, such as Consolidation in region 1 and Enlarged Mediastinum in region 2, offering only non-specific descriptions and contradicting the clinical facts.

To address these issues, we propose a Multi-Granularity Knowledge Guided RAG framework for radiology reports. Specifically, we build a multi-granularity knowledge guided multimodal retriever. We use multi-granularity intra-report similarities to guide cross-modal image-text alignment, enabling the retriever to surpass superficial similarity and precisely recall cases with high medical-factual consistency. Finally, we construct a knowledge graph from the retrieved cases and use an attention mechanism to enhance key pathological visual features, ultimately generating clinically more accurate and detailed reports.

## 2 Methodology

### 2.1 Framework

As shown in Fig. 2, we propose a multi-granularity knowledge-guided RAG framework for radiology reports, which consists of: (a) *Multi-Granularity Knowledge Guided Multimodal Retrieval* trains a multimodal retriever to align image-text representations using multi-granular knowledge constraints, retrieving  $K$  relevant reports from a FAISS index. (b) *Cross-Modal Knowledge-Augmented Report Generation* utilizes cross-attention to integrate the original image features with a knowledge graph derived from these retrieved reports, yielding enhanced visual features that guide a large model to generate a factually consistent report.

### 2.2 Multi-Granularity Knowledge Guided Multimodal Retrieval

Given a batch of image-report pairs  $\{(I_i, R_i)\}_{i=1}^N$ , we augment the standard InfoNCE loss with text-based, multi-granularity semantic

similarities to refine cross-modal alignment. Specifically, the image and text encoders produce features  $v_i$  and  $t_i$ . We then derive the contrastive features  $\hat{v}_i$  and  $\hat{t}_i$  by projecting their global representations and applying  $L_2$ -normalization.

**Image-Report Contrastive Learning.** We first apply the standard InfoNCE loss  $L_{CL}$  for basic cross-modal alignment, which is the average of the image-to-text and text-to-image losses  $L_{i2t}$  and  $L_{t2i}$ :

$$\mathcal{L}_{CL} = \frac{1}{2} (\mathcal{L}_{i2t} + \mathcal{L}_{t2i}) = -\frac{1}{2N} \sum_{i=1}^N \left[ \log \frac{\exp(\cos(\hat{v}_i, \hat{t}_i)/\tau)}{\sum_{j=1}^N \exp(\cos(\hat{v}_i, \hat{t}_j)/\tau)} + \log \frac{\exp(\cos(\hat{t}_i, \hat{v}_i)/\tau)}{\sum_{j=1}^N \exp(\cos(\hat{t}_i, \hat{v}_j)/\tau)} \right]. \quad (1)$$

where  $\tau$  is a learnable temperature hyperparameter.

However,  $\mathcal{L}_{CL}$  only models coarse-grained image-report correspondences, ignoring the hierarchical semantic relations among reports, which results in representations that fail to reflect detailed clinical semantics. Therefore, we propose  $\mathcal{L}_{MGK}$ , a multi-granularity knowledge guided loss that leverages this intra-modal knowledge. Its core idea is to minimize the divergence between the cross-modal similarity distribution  $P$  and a text-derived soft-label distribution  $Q$ . This distribution  $P$  includes both the image-to-text  $P_{ij}^{i2t}$  and text-to-image  $P_{ij}^{t2i}$  directions:

$$P_{ij}^{i2t} = \frac{\exp(\cos(\hat{v}_i, \hat{t}_j)/\tau)}{\sum_{k=1}^N \exp(\cos(\hat{v}_i, \hat{t}_k)/\tau)}, \quad (2)$$

$$P_{ij}^{t2i} = \frac{\exp(\cos(\hat{t}_i, \hat{v}_j)/\tau)}{\sum_{k=1}^N \exp(\cos(\hat{t}_i, \hat{v}_k)/\tau)}. \quad (3)$$

**Report-Report Knowledge Constraint Contrastive Learning.** To compute the three granular soft-label distributions  $Q_r, Q_l, Q_f$  and their corresponding loss terms  $\mathcal{L}_r, \mathcal{L}_l, \mathcal{L}_f$ , we first define three distinct similarity matrices  $S_r, S_l, S_f$ :

**Global Semantic Similarity  $S_r$ .** We compute the cosine similarity matrix  $S_r$  from the text encoder's output  $\hat{t}$ :

$$S_r = \hat{t} \cdot \hat{t}^T, \quad (4)$$

**Pathological Label Similarity  $S_l$ .** To capture key clinical diagnoses, we use CheXpert to extract 14 pathological labels, which are

converted into multi-label 0/1 vectors and then  $L_2$ -normalized as  $\hat{L} \in \mathbb{R}^{N \times d_l}$ . The similarity matrix  $S_l$  is defined as:

$$S_l = \hat{L} \cdot \hat{L}^T, \quad (5)$$

**Clinical Fact Similarity  $S_f$ .** To capture fine-grained clinical details such as lesion location and severity, we use RadGraph to extract triplets, map them to SNOMED CT to reduce terminological inconsistencies, and compute the F1RadGraph score  $S_f[i, j]$  between the triplet sets  $G(\cdot)$  of reports  $R_i$  and  $R_j$ :

$$S_f[i, j] = \frac{2 |G(R_i) \cap G(R_j)|}{|G(R_i)| + |G(R_j)|}, \quad (6)$$

to convert each similarity matrix  $S_k$  into its respective soft-label distribution  $Q_k$ , we first apply a thresholding operation, followed by row-wise normalization. Therefore, the image-to-text constraint loss  $\mathcal{L}_k(I, T)$  is:

$$\mathcal{L}_k(I, T) = \frac{1}{N} \sum_{i=1}^N \text{KL}(Q_k[i] \parallel P_i^{i2t}), \quad (7)$$

where  $k \in \{r, l, f\}$  and  $P_i^{i2t}$  is the  $i$ -th row of  $P^{i2t}$ . Similarly, we compute the text-to-image constraint loss  $\mathcal{L}_k(T, I)$ .

The total image-to-text knowledge constraint loss  $\mathcal{L}_{MGK}^{i2t}$  is the weighted sum of the three granular losses:

$$\mathcal{L}_{MGK}^{i2t} = \lambda_r \mathcal{L}_r(I, T) + \lambda_l \mathcal{L}_l(I, T) + \lambda_f \mathcal{L}_f(I, T). \quad (8)$$

Similarly, we compute the total text-to-image constraint loss  $\mathcal{L}_{MGK}^{t2i}$ . The final training objective is:  $\mathcal{L}_{\text{total}}$  is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{CL} + \alpha (\mathcal{L}_{MGK}^{i2t} + \mathcal{L}_{MGK}^{t2i}). \quad (9)$$

We set  $\lambda_r = \lambda_l = \lambda_f = \frac{1}{3}$  and  $\alpha = 0.5$  by default. Minimizing this joint loss improves the factual accuracy of report retrieval.

## 2.3 Cross-Modal Knowledge-Augmented Report Generation

To improve the factual accuracy of generated reports, we fuse retrieved medical knowledge with visual features. Given an input image  $i$ , we extract visual features  $v_i$  with the encoder and retrieve Top-K reports  $T_i^*$  using our multi-granularity knowledge guided retriever. We then employ RadGraph to extract knowledge triplets  $K_i$  from these retrieved reports. These triplets are structured into an instance-specific knowledge graph  $G = (f_N, A)$  with node features  $f_N$  and adjacency matrix  $A$ . We encode structural knowledge in graph  $G$  using a relational self-attention layer and an FFN, obtaining the graph knowledge representation  $f_{KG}$ :

$$f_{KG} = \text{FFN}(\text{RSA}(f_N, A)). \quad (10)$$

Using  $v_i$  as the Query and  $f_{KG}$  as the Key and Value, we inject graph knowledge via cross-attention, yielding  $e$ . This intermediate feature  $e$  then sequentially passes through a residual connection to  $v_i$ , a LayerNorm, an FFN, and a final residual-plus-LayerNorm operation to produce the knowledge-enhanced  $v_i^*$ :

$$v_i^* = \text{LN}(\text{FFN}(e) + v_i). \quad (11)$$

We adopt a parameter-frozen Llama3.1-8B as the decoder. We feed both the projected visual representation  $v_i^*$  and a prompt  $P$ , constructed from an instruction and the retrieved reports  $\{T\}_i^*$ , into

the LLM, the language modeling loss is defined as follows:

$$\mathcal{L}_{LLM} = - \sum_{t=1}^T \log p(r_t^p | v_i^*, P, r_{1:t-1}^p). \quad (12)$$

Predicted report  $\{r_1^p, \dots, r_T^p\}$ , where  $r_t^p \in \mathbb{V}$  is the token at step  $t$ .

## 3 Experiments

### 3.1 Experimental Settings

**Datasets.** MIMIC-CXR [9] is a large-scale public dataset containing 227,835 radiology reports. We adhere to the official data splits for fair evaluation and comparison. IU X-Ray [5] comprises 7,470 images and 3,955 reports, which we split into training, validation, and test sets in a 7:2:1.

**Evaluation Metrics.** Following previous studies [15], we adopt two categories of metrics to evaluate our method and the baselines: Natural Language Generation (NLG) and Clinical Efficacy (CE). The NLG metrics include ROUGE-L [4] and BERTScore [18]. The CE metrics include F1CheXbert [19] and F1RadGraph [7].

**Implementation Details.** We use the frontal view of each radiology study, a Swin Transformer pre-trained on ImageNet as the visual encoder, BioClinicalBERT as the text encoder, and LLaMA3.1-8B as the decoder. We train on MIMIC-CXR for 10 epochs using the AdamW optimizer with a batch size of 32 and a learning rate of  $5 \times 10^{-6}$ . For top- $k$  retrieval, we set  $k = 3$ .

### 3.2 Comparison with State-of-the-art Methods

As shown in Table 1, retrieval-based methods clearly outperform pure generation baselines on clinical metrics. On MIMIC-CXR, our MGK-RAG further improves F1CheXbert from 0.602 to 0.614 and F1RadGraph from 0.257 to 0.262 over the best prior retrieval model FactMM-RAG, while keeping ROUGE-L and BERTScore comparable. As shown in Table 2, MGK-RAG on IU X-Ray likewise achieves the best F1CheXbert and F1RadGraph and competitive ROUGE-L and BERTScore, indicating that our gains in clinical factuality generalize without degrading NLG quality.

**Table 1: Comparison results on MIMIC-CXR.**

| Type       | Model           | F1CheXbert   | F1RadGraph   | ROUGE-L      | BERTScore    |
|------------|-----------------|--------------|--------------|--------------|--------------|
| Generation | LLaVA-Med [10]  | 0.166        | 0.048        | 0.170        | 0.261        |
|            | MiniGPT-Med [2] | 0.214        | 0.147        | 0.271        | 0.361        |
|            | XrayGPT [16]    | 0.291        | 0.165        | 0.285        | 0.368        |
|            | R2GenGPT [17]   | 0.389        | 0.187        | 0.297        | 0.415        |
|            | META-CXR [6]    | 0.428        | 0.224        | 0.280        | 0.426        |
| Retrieval  | LaB-RAG [14]    | 0.466        | 0.183        | 0.204        | <b>0.861</b> |
|            | FactMM-RAG [15] | <u>0.602</u> | <u>0.257</u> | <b>0.307</b> | <u>0.561</u> |
|            | <b>Ours</b>     | <b>0.614</b> | <b>0.262</b> | <u>0.299</u> | 0.557        |

### 3.3 Ablation Study

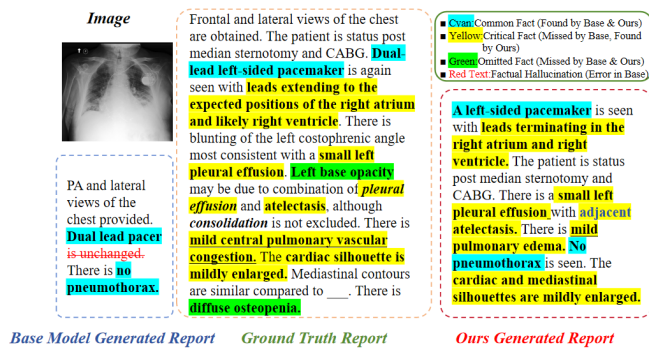
As shown in Table 3, we perform ablation experiments on MIMIC-CXR to quantify the contribution of each component in MGK-RAG. The framework comprises a Multi-Granularity Knowledge Guided Multimodal Retrieval module with intra-modal constraints R, L and F, and a Cross-Modal Knowledge-Augmented Report Generation module E. The base model, which uses only the visual encoder and the LLM decoder without retrieval, attains 0.505/0.239 on F1CheXbert/F1RadGraph and 0.297/0.551 on ROUGE-L/BERTScore.

**Table 2: Comparison results on IU X-Ray.**

| Type       | Model           | F1CheXbert   | F1RadGraph   | ROUGE-L      | BERTScore    |
|------------|-----------------|--------------|--------------|--------------|--------------|
| Generation | LLaVA-Med [10]  | 0.140        | 0.075        | 0.160        | 0.261        |
|            | MiniGPT-Med [2] | 0.489        | <u>0.310</u> | <b>0.345</b> | 0.472        |
| Retrieval  | LaB-RAG [14]    | <u>0.536</u> | 0.228        | 0.219        | <b>0.865</b> |
|            | <b>Ours</b>     | <b>0.543</b> | <b>0.314</b> | <u>0.273</u> | <u>0.524</u> |

**Table 3: Model performance with different designed modules.**

| R | L | F | E | F1CheXbert   | F1RadGraph   | ROUGE-L      | BERTScore    |
|---|---|---|---|--------------|--------------|--------------|--------------|
|   |   |   |   | 0.505        | 0.239        | 0.297        | 0.551        |
| ✓ |   |   | ✓ | 0.547        | 0.245        | 0.297        | 0.555        |
| ✓ | ✓ |   | ✓ | <u>0.599</u> | <u>0.259</u> | <u>0.297</u> | <u>0.556</u> |
| ✓ | ✓ | ✓ | ✓ | <b>0.614</b> | <b>0.262</b> | <b>0.299</b> | <b>0.557</b> |

**Figure 3: Qualitative examples on MIMIC-CXR.**

Introducing the report-level constraint R, training a retriever and using module E to inject retrieved knowledge into the visual representation improves the clinical metrics to 0.547/0.245 with comparable NLG scores; adding the label-level constraint L further raises them to 0.599/0.259. The full model, which additionally leverages the fact-level constraint F, achieves 0.614/0.262 and the best ROUGE-L and BERTScore, demonstrating that each component of MGK-RAG contributes to generating more factually accurate radiology reports.

### 3.4 Case Study

Fig. 3 presents an example of our qualitative comparison results. The baseline report on the left is factually poor: it identifies only common facts, such as *Dual lead pacer* and *no pneumothorax*, suffers from the factual hallucination *is unchanged*, and critically omits all key positive findings, such as *small left pleural effusion*, *atelectasis*, and *cardiac silhouette is mildly enlarged*. In contrast, our model on the right demonstrates superior factual completeness.

## 4 Conclusion

We propose a Multi-Granularity Knowledge Guided RAG framework to mitigate factual inconsistencies in radiology report generation. Our framework guides contrastive learning via report, label, and fact-level semantic similarities to enhance factual retrieval, and fuses retrieved reports with images to obtain knowledge-augmented visual representations. The method achieves leading performance on the MIMIC-CXR and IU X-Ray benchmarks, generating clinically accurate and reliable reports.

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